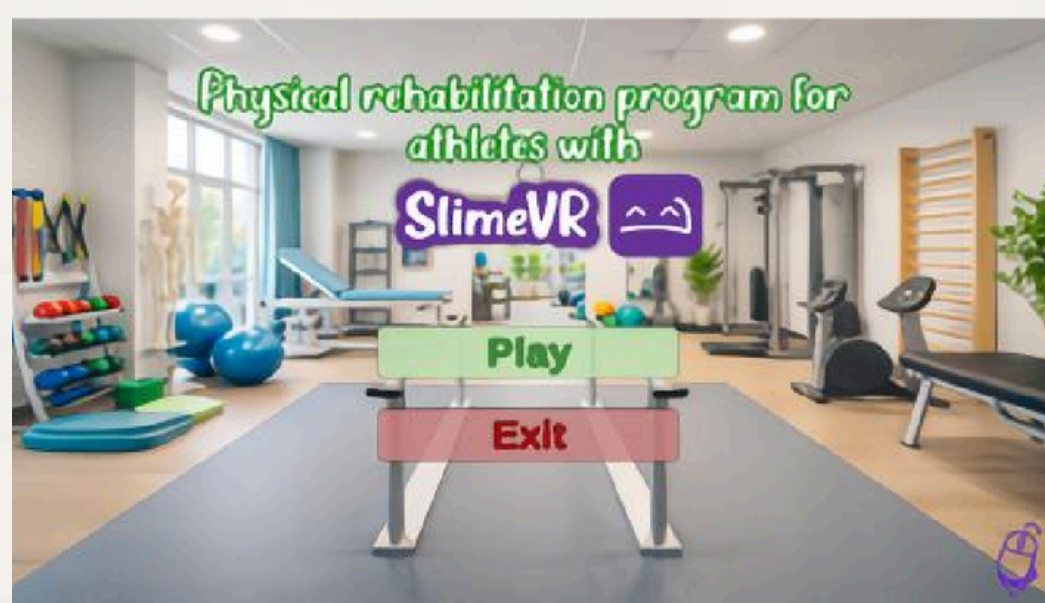


PROPOSAL FOR PHYSICAL REHABILITATION

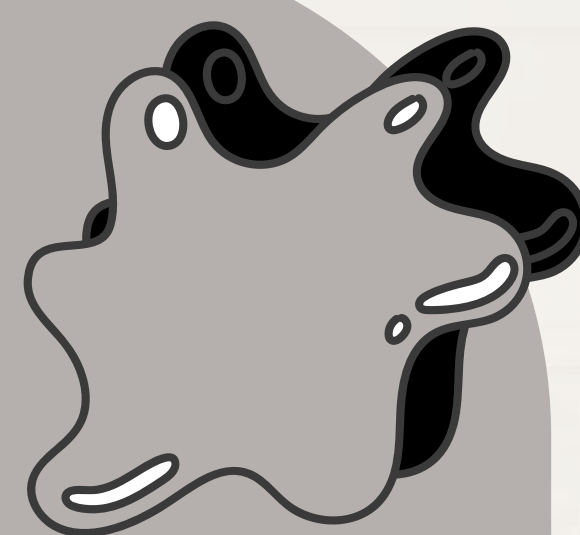
based on virtual reality using motion sensors: Inertial Motion Units (IMU) applied to university athletes.

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Abstract

The system is designed to guide and monitor shoulder and knee exercises through real-time body tracking and dynamic repetition goals.



This study does not evaluate clinical effectiveness but provides an engineering validation of IMU-based tracking systems as a viable alternative to camera-based approaches for interactive rehabilitation support.

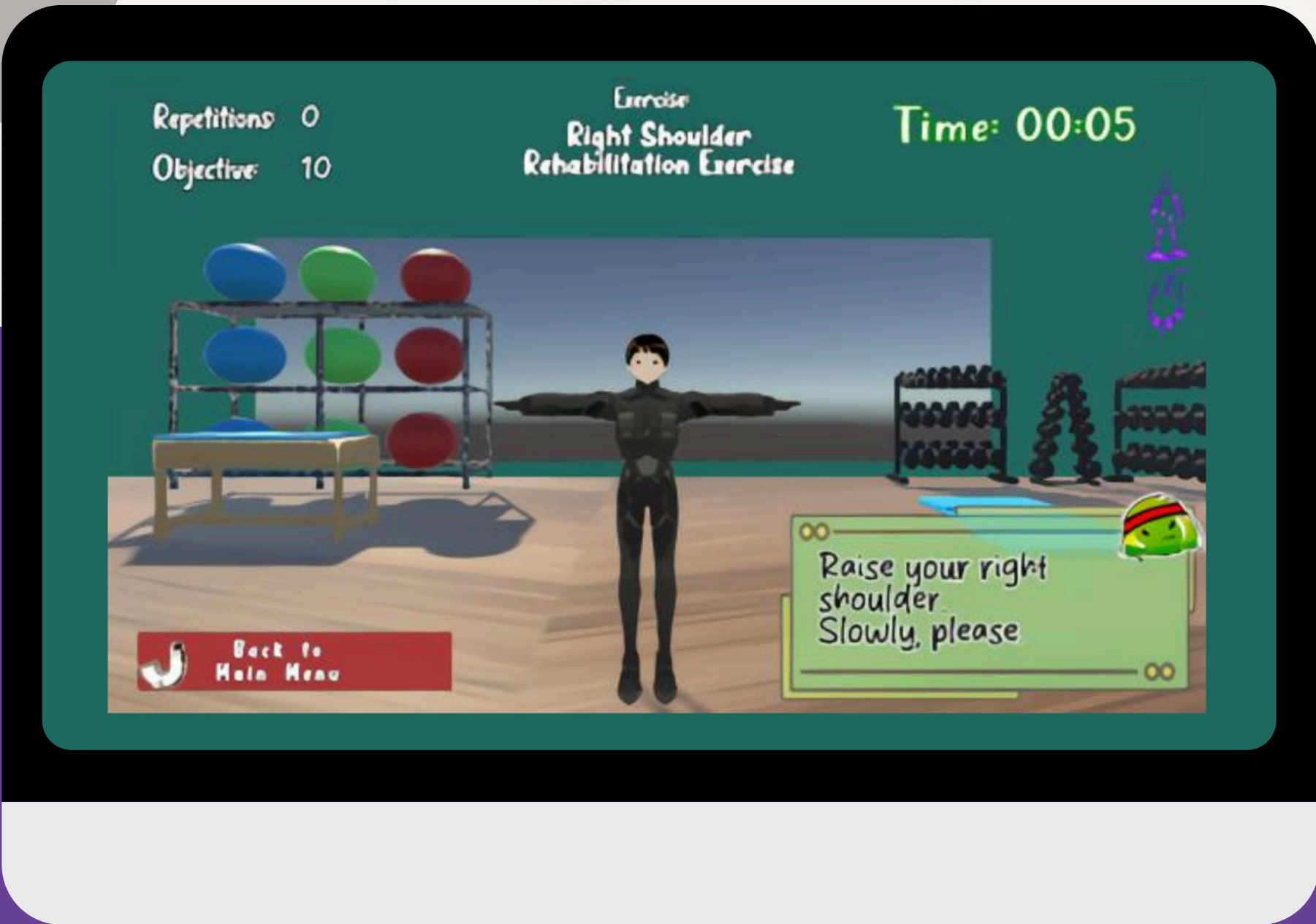
Introductions

Physical rehabilitation is a fundamental component in the functional recovery of patients with musculoskeletal injuries and motor disorders. However, traditional therapeutic processes often face problems with patient adherence and motivation, mainly due to the repetitive and monotonous nature of the pre-scribed exercises.

The results of this study aim to provide evidence on the feasibility of using inertial movement units (IMUs) in interactive rehabilitation applications, as well as to provide a reference for the design and evaluation of future therapeutic systems based on motion tracking .

Methodology

This study involved the design, implementation, and evaluation of an interactive, motion-based rehabilitation system developed in the Unity3D Game Engine and integrated with inertial sensors. The methodology was divided into two main components: The development of the interactive application and its bodytracking architecture. The experimental evaluation through user testing and structured surveys.



Results

The usability evaluation revealed very positive perceptions among participants. Participants rated the clarity of the exercise objectives with an average score of 9.33, while the ease of understanding the required movements reached an average of 9.60. Similarly, the visual organization of the interface received an average score of 9.56, suggesting that the system's design is intuitive for user interaction.

Conclusions

Based on the observations obtained from the survey ratings, we can identify the following areas for improvement in these statistics:

- **Error detection (8.67):** The system could be im-proved in alerting users when a movement is incorrect.
- **Physical exertion (8.67):** Some users felt the difficulty could be better adjusted.
- **Technical errors (8.77):** Minor stability issues were observed.
- **Environmental realism (8.87):** The virtual environment could be even more immersive
Future work will focus on evaluating the system with individuals undergoing real-world rehabilitation to assess its potential clinical impact.

